



MARKET MONITOR

Contents

Feature article –China's revised production estimates	1
World supply-demand outlook	3
Crop monitor	5
Policy developments	8
International prices	9
Futures markets	11
Market indicators	12
Monthly US ethanol update	14
Fertilizer outlook	15
Ocean freight market update	16
Explanatory notes	17

No. 64 – December 2018

This year has been eventful in many ways for most agricultural commodities; beyond the usual supply and demand factors shaping markets there has been an increasing level of uncertainty stemming from unexpected shifts in trade policies. Among the AMIS crops, maize and soybeans have been at the forefront of these policy changes, while global wheat and rice markets have been little affected. The other important feature, also central to the work of AMIS, is the recent revelation of drastically higher cereal production levels in China with potentially important implications, which is the main focus of this month's editorial.

Markets at a glance

	From previous forecast	From previous season
Wheat	■	▼
Maize	■	▼
Rice	■	▲
Soybeans	▲	▲

▲ Easing ■ Neutral ▼ Tightening

The **Market Monitor** is a product of the Agricultural Market Information System (AMIS). It covers international markets for wheat, maize, rice and soybeans, giving a synopsis of major market developments and the policy and other market drivers behind them. The analysis is a collective assessment of the market situation and outlook by ten international organizations and entities that form the AMIS Secretariat.

Visit us at: www.amis-outlook.org

China's new grain production estimates

China¹ has revised its grain production estimates. This is not an unusual move, neither for China nor for any other country. The revisions emanated from the 2017 census, the results of which have recently been released by the National Bureau of Statistics (NBS). Unusual, however, is the extent of the revisions and the fact that they go all the way back to the last census, which took place in 2007. It is also unusual that all revisions go only in one direction: upwards. In every single year since 2007, grain production is now higher than hitherto reported. Cumulatively, the upward revisions for maize alone amount to 266 million tonnes. For all cereals combined, the upwards revisions add up to 312 million tonnes. The sheer magnitude of this revision confronts market analysts with the challenge of how to accommodate the cumulative production increase in China's grain balances, and more specifically, how to distribute the extra grain over the various forms of utilization. Has there been more feed, food, fuel, waste or should we simply assume all that grain has ended up in the country's suspected already overfilled granaries? Among the various end-users of the extra production, two candidates stand out: feed and stocks.

How much more feed use is justified?

While the census provides no immediate guidance on how to change the feed use allocations, it offers some useful data points to narrow down the estimates. Two pieces of information are of particular interest: The first is the revision of China's livestock herd, particularly the number and type of livestock. The second is the change in farm types and sizes and hence the likely changes in feeding *intensities* and feeding *efficiencies*.

Even a cursory look at the census results for the livestock herds and their composition suggests a *smaller* overall herd size and only a very moderate shift from ruminants to more grain-intensive livestock, i.e. pigs, waterfowl and aquaculture. This makes it difficult to justify a sharp increase in feed allocations. Similarly, there is no significant potential to assume higher feeding *intensities*, i.e. a higher use of compound and concentrate feeds that comes with structural change. The shifts from backyard producers, to specialised family farmers, to industrial operations were rather modest. What is more, they should have been offset by higher feeding *efficiencies*, i.e. a lower amount of feed needed per unit of livestock output in a given system.

Where has the extra grain gone to?

Without a solid justification for higher feed use, most of the additional grain could only be assumed to have ended up in stocks. While convenient from an accounting perspective, relegating the surplus to stocks poses a whole range of new questions. The most obvious are: where are these stocks, and who holds them? More specifically, are these stocks held as strategic reserves by the government, by farmers or by processors as additional pipeline stocks? How plausible is it to assume that they are on-farm stocks and would China's smallholder farmers really have the requisite storage capacity to carry them – together with the new crop – into the next crop year? No doubt, the smallholder farmers have traditionally held large grain reserves; they did so to have a store of value, an insurance against a bad crop, a means to ensure food security in China's once underdeveloped rural economy. But China's rural hinterland has undergone significant structural change, the number of smallholder farmers has declined, rapidly and massively. By one estimate it has fallen by half over the past 20 years and there is no reason to assume that the remaining farmers still need all that grain as a store of value. If, on the other hand, the extra grain is held by the government, their grain elevators should be brimming over, literally.

Why may the balancing approach no longer be an option?

The most fundamental question arising from these revisions, however, is whether the practice of using stocks as a balancing item, or a residual, is still a defensible approach to inform an international surveillance system such as G20-AMIS. Relegating imbalances to stocks means that the measurement errors inherent in all other elements of the balance have to be absorbed by the estimate for stocks. This may be acceptable, if we could have accurate data on all those elements or if there was a good reason to assume that the measurement errors for all other elements cancel out. But that is not the case. In fact, the errors tend to be additive and create a vastly distorted stock estimate. That poses the next question:

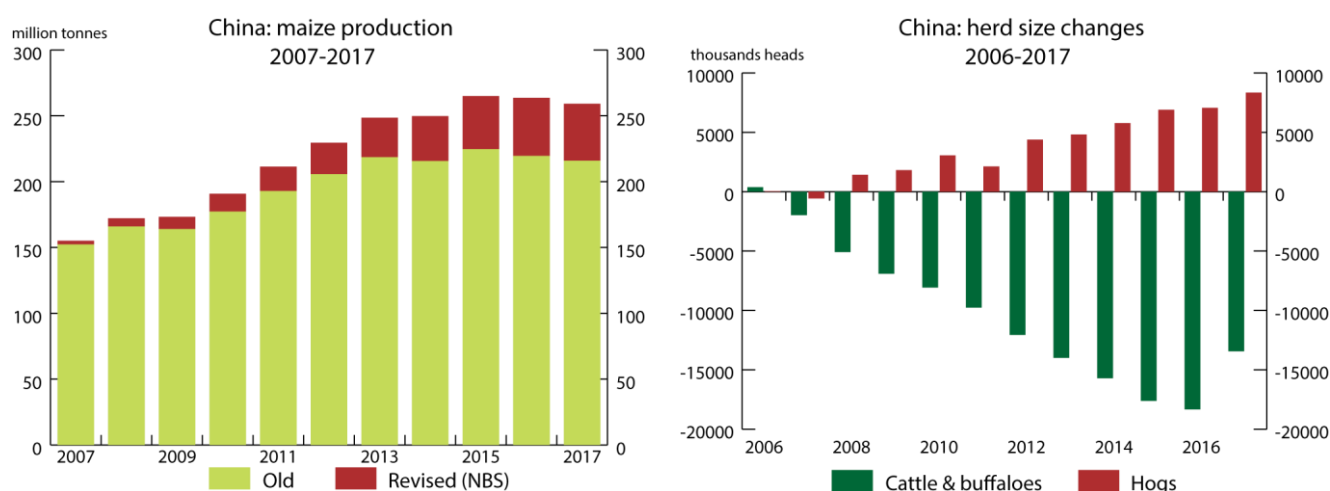
¹ For the purpose of this article "China" refers only to mainland China

Can the international community afford to have low-quality stock estimates?

Stocks are arguably still the most critical element to assess the resilience of international food markets. This is why improving the quality of data on grain stocks has played such a central role to AMIS. At a global level, stocks and stock-to-use ratios allow analysts to gauge the capacity to buffer against exogenous shocks, and understanding the responsiveness of stocks to market signals is critical to capture the vulnerability of the world's food system. However, for stocks to play this role, the estimates must be accurate, they must be reliable. It is hard to accept, particularly for a player as important as China, that we obtain stock estimates as a residual in the supply and demand balance. What is required instead is to shift to a survey-based system to estimate the level of inventories at all stages of the agricultural supply chain. Having accurate stock estimates is in fact a global public good the world cannot forego; it is also at the heart of the AMIS mandate. Failing to produce more accurate global stock estimates could expose the world food system to unexpected price swings as those experienced ten years ago, if not even to a new global food crisis.

While the recent census has put the spotlight on China's grain estimates, similar issues bedevil the balances of many other AMIS countries. The long-term solution can lie in more and better survey-based estimates. But in the short-run, some other options to produce better balances will be pursued and, importantly, for all countries. These would include ensuring that feed estimates are in line with herd sizes, composition, feeding intensities and efficiencies. It also involves reviewing the amount of grains and oilseeds allocated to industrial use and juxtapose them to industrial output such as biofuels, paints or detergents. Or to simply review the amounts allocated to waste and relate them to levels of disappearance, ensuring that the resulting ratios remain within intuitive and corroborative ranges.

This Market Monitor offers world estimates with and without China. The sole purpose of which is to illustrate the magnitude of the necessary changes, not to attribute responsibilities or problems.



World supply-demand outlook

World Balances, with and without China*

(in million tonnes)

- **Wheat** production forecast lowered slightly, mostly on downward revisions in the Russian Federation and Turkey.
- Utilization in 2018/19 lowered marginally following a further downgrading of prospects for feed use of wheat.
- Trade in 2018/19 (July/June) lowered due to smaller expected imports in north Africa; exports from Argentina and the Russian Federation are trimmed since last month.
- Stocks (ending 2019) almost unchanged m/m but 4.4 percent below their record high opening levels, driven by drawdowns among the major wheat exporters.

WHEAT	FAO-AMIS			USDA		IGC	
	2017/18 est.	2018/19 f'cast		2017/18 est.	2018/19 f'cast	2017/18 est.	2018/19 f'cast
		01-Nov	06-Dec		08-Nov		22-Nov
Production	760.3	727.9	725.1	763.1	733.5	767.1	728.5
	627.3	599.9	597.1	628.7	601.0	628.9	593.8
Supply	1 015.4	1 004.7	1 001.9	1 024.1	1 012.5	1 011.5	1 001.1
	793.6	771.8	769.0	774.8	748.8	772.2	744.5
Utilization	738.0	740.1	739.6	745.1	745.8	738.9	739.2
	617.4	619.4	618.9	624.1	622.8	618.6	612.4
Trade	176.2	173.2	172.5	181.3	178.8	176.4	171.2
	172.5	169.2	168.5	177.3	174.8	172.7	167.2
Stocks	276.7	264.4	264.5	279.0	266.7	272.5	261.8
	171.9	148.3	148.5	147.7	123.1	154.7	133.1

- **Maize** production forecast down marginally m/m with lower anticipated harvests in the Russian Federation and the US more than offsetting a significant upward revision in Ukraine.
- Utilization in 2018/19 scaled down on cuts in feed use in the US and the Russian Federation more than offsetting a raise in the EU.
- Trade in 2018/19 (July/June) lifted, supported by stronger import demand by several countries; exports from Ukraine much higher than earlier anticipated on the back of record production.
- Stocks (ending in 2019) unchanged from last month and still 14 percent below their record opening levels; the biggest decreases are expected in China, South America and the US.

MAIZE	2017/18	2018/19		2017/18	2018/19	2017/18	2018/19
Production	1 093.2	1 068.9	1 068.4	1 076.2	1 099.0	1 047.4	1 073.4
	877.3	852.4	851.9	941.9	843.0	831.5	853.5
Supply	1 393.1	1 379.9	1 379.5	1 426.5	1 439.9	1 377.8	1 378.7
	1 019.1	1 016.8	1 040.4	944.4	961.3	957.4	965.6
Utilization	1 073.0	1 109.8	1 107.9	1 085.6	1 132.4	1 072.5	1 112.4
	842.4	865.2	863.3	822.6	856.4	840.8	864.5
Trade	155.0	155.8	157.2	147.1	163.0	151.6	158.7
	150.7	151.3	152.2	143.7	158.0	147.5	154.5
Stocks	311.1	267.4	267.5	340.9	307.5	305.3	266.3
	164.5	144.9	145.0	118.4	100.0	116.6	101.2

- **Rice** production in 2018 only fractionally changed m/m, as a reduction for Madagascar is largely outweighed by improved prospects for a few Latin American countries.
- Utilization forecast for 2018/19 trimmed, reflecting less buoyant use prospects for Brazil and African countries, namely Nigeria.
- Trade in 2019 forecast to edge down by 1.4 percent y/y, as lower imports by Bangladesh, China and Indonesia more than offset larger purchases by Egypt, Nigeria and the Philippines
- Stocks (2018/19 carry-out) still seen expanding by 2.7 percent to an all-time high.

RICE (milled)	2017/18	2018/19		2017/18	2018/19	2017/18	2018/19
Production	506.3	513.0	512.9	494.3	490.7	490.5	490.6
	363.4	371.7	371.6	345.4	347.1	345.5	344.6
Supply	674.3	685.0	685.0	643.3	651.4	613.9	617.4
	432.7	440.7	440.7	395.9	398.8	392.5	395.9
Utilization	503.8	509.2	508.8	482.5	488.4	487.2	492.7
	360.6	365.5	365.1	340.0	345.6	342.2	346.7
Trade	47.9	47.3	47.2	48.7	49.2	47.6	48.6
	42.6	42.3	42.2	43.7	44.2	42.4	43.6
Stocks	172.1	176.6	176.7	160.7	163.0	126.8	124.4
	69.1	72.5	72.6	51.7	50.0	52.1	50.8

- **Soybean** 2018/19 production trimmed marginally, with reductions in the US and the Russian Federation only partially offset by higher forecasts for Argentina.
- Utilization in 2018/19 reduced somewhat, mainly reflecting lower than earlier anticipated forecasts for Brazil, following a brisk export path, as well as China, tied to weakening demand for meal.
- Trade forecast for 2018/19 revised down slightly, underpinned by smaller than earlier expected imports by China.
- Stocks (2018/19 carry-out) raised further, with upward revision in the US outweighing reductions in other countries. Global inventories now forecast to climb by more than one-third y/y.

SOYBEANS	2017/18	2018/19		2017/18	2018/19	2017/18	2018/19
Production	341.5	371.1	369.0	338.6	367.5	340.1	367.4
	326.2	355.3	353.2	323.4	351.5	325.1	351.6
Supply	390.4	412.2	410.1	436.0	467.2	384.0	407.1
	360.3	383.0	380.9	400.1	427.7	257.3	284.5
Utilization	348.3	356.8	354.7	336.8	351.9	344.2	356.3
	238.1	247.3	246.1	230.5	242.3	236.4	247.1
Trade	153.2	155.6	154.8	153.2	155.4	152.4	152.1
	59.1	64.3	64.5	59.6	62.3	60.1	64.1
Stocks	41.1	54.2	55.3	99.7	112.1	39.7	50.7
	27.7	43.2	44.8	76.2	92.2	20.9	37.4



World Balances

* Data shown in the second rows refer to world aggregates without China; world trade data refer to exports and world trade without China excludes exports to China.

To review and compare data, by country and commodity, across three main sources, go to <http://statistics.amis-outlook.org/data/index.html#COMPARE>

Estimates and forecasts may differ across sources for many reasons, including different methodologies. **For more information see Explanatory notes on the last page of this report.**

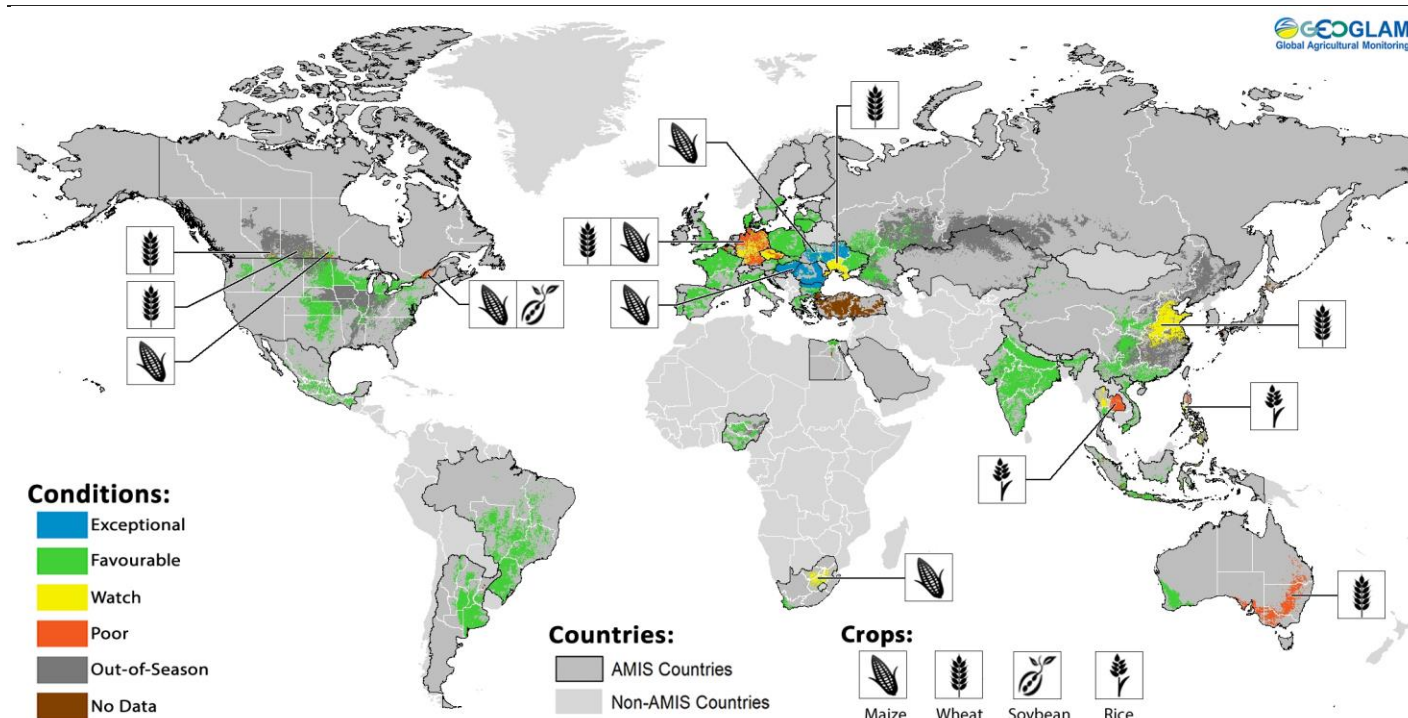
Summary of revisions to FAO-AMIS monthly forecasts for 2018/19

in thousand tonnes

[illegible]

Crop monitor

Crop conditions in AMIS countries (as of 28 November)



Crop condition map synthesizing information for all four AMIS crops as of 28 November. Crop conditions over the main growing areas for wheat, maize, rice, and soybean are based on a combination of national and regional crop analyst inputs along with earth observation data. **Only crops that are in other-than-favourable conditions are displayed on the map with their crop symbol.**

Conditions at a glance

Wheat - In the southern hemisphere, winter wheat conditions are favourable entering into harvest in Argentina and South Africa with the exception of Australia. In the northern hemisphere, winter wheat has emerged and entered dormancy in many places. Several areas of concern due to dry conditions remain in parts of northern Europe, southern Ukraine, and in the North China Plain.

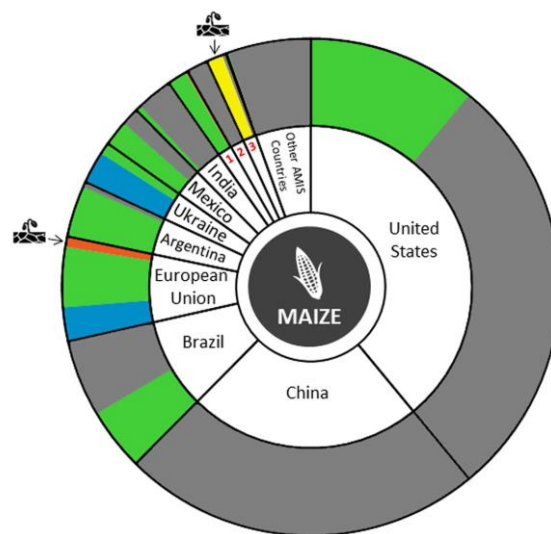
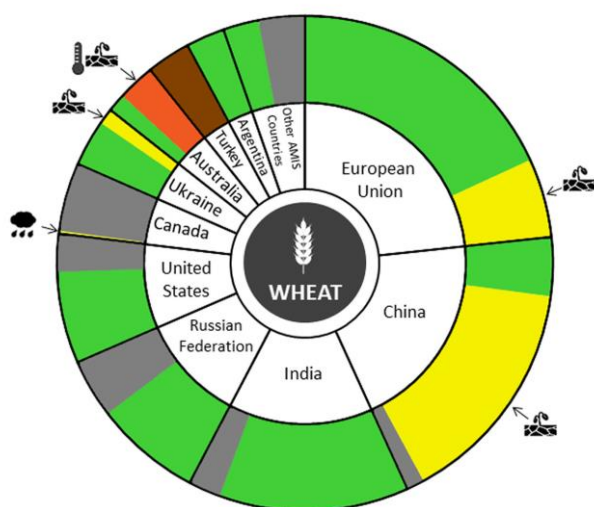
Maize - In the northern hemisphere, harvest is primarily complete. In the southern hemisphere, sowing is progressing in Brazil and Argentina under favourable conditions, while there are dry conditions in South Africa.

Rice - In India, Kharif rice harvest is progressing and Rabi rice sowing is beginning. In Southeast Asia, harvest of wet-season rice is nearing the end in the northern countries with poor areas in the Philippines and Thailand, while sowing of wet-season rice is advancing in Indonesia.

Soybean - In the northern hemisphere, harvest is now complete with India and Canada wrapping up. In the southern hemisphere, sowing is ongoing in Brazil and Argentina under favourable conditions.

Weak to moderate El Niño expected in the first half of 2019

A weak to moderate El Niño is expected to form and continue through Northern Hemisphere 2018/19 winter until spring (~80 percent chance through February; 55 percent chance through March-May). Associated with the potential development of this El Niño event between December and February are increased chances of above normal rainfall in parts of Tanzania, Central Asia, the southern US, Mexico, and southeastern South America. Drier than normal conditions are anticipated for the Indo-Pacific region, including parts of southeast Asia, Indonesia, and Australia, and for parts of Central America, the Caribbean, northern Brazil and Southern Africa.

Conditions:**Drivers:**Canada¹, Russian Federation², South Africa³**Wheat**

In the **EU**, winter wheat conditions are mostly favourable through there is concern developing over areas of persistent dry soil conditions, most notably in Germany and the Czech Republic, limiting emergence and early crop development. In **Ukraine**, winter wheat is now in dormancy under generally favourable conditions with some areas of underdeveloped crops due to dry conditions in the southern and eastern regions. In the **Russian Federation**, winter wheat established under favourable conditions and the crop is now dormant. In **China**, winter wheat conditions are mixed with dry conditions developing after sowing and emergence in the North China Plain. In **India**, sowing of winter wheat is progressing under favourable conditions. In the **US**, winter wheat sowing and emergence is on time and under favourable conditions. In **Canada**, winter wheat sowing is mostly complete under mixed conditions in the western Prairies due to adverse sowing conditions in the late fall. A reduction in total sown area is expected. In **Australia**, harvest of winter wheat is well underway with production forecast to decrease significantly year-on-year. In **Argentina**, harvest is ongoing in the north under favourable conditions with an overall increase in production expected owing to an increase in total sown area and a slight increase in yields.

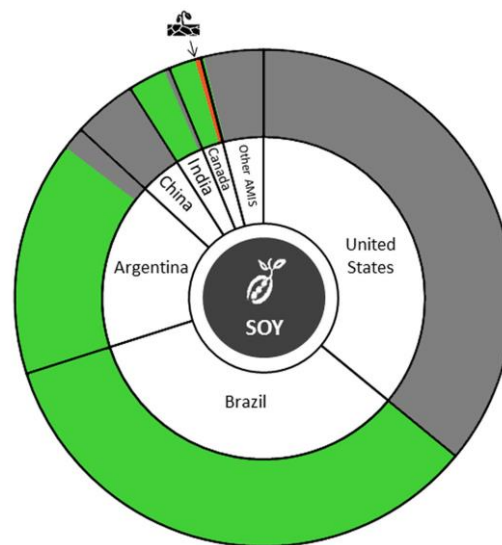
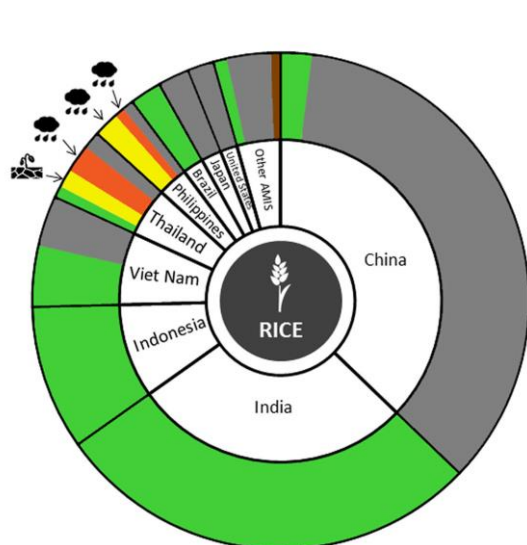
Maize

In the **US**, harvest is almost complete with a bumper crop in many parts of the country. In **Canada**, harvest is closing under generally favourable conditions, but with some areas of variable yields due to dry conditions during the season. In **Mexico**, harvest of the spring-summer crop is ongoing under favourable conditions. In the **EU**, harvest is almost complete with overall EU yields remaining above the five-year average owing to favourable to exceptional conditions in southern Europe. In **Ukraine**, harvest is wrapping up with record yields in the central and western regions. In **India**, sowing of the Rabi crop is progressing under favourable conditions. In **Brazil**, the spring-planted crop is developing under favourable conditions in the main producing regions, an increase in total sown area expected. In **Argentina**, sowing is complete for the spring-planted crop under favourable conditions with recent rainfall improving soil moisture conditions. Sowing of the summer-planted crop is just beginning. In **South Africa**, sowing conditions are mixed with only the far eastern areas receiving sufficient rainfall.



Pie chart description: Each slice represents a country's share of total AMIS production (5-year average), with the main producing countries (95 percent of production) shown individually and the remaining 5 percent grouped into the "Other AMIS Countries" category. Sections within each country are weighted by the sub-national production statistics (5-year average) of the respective country and accounts for multiple cropping seasons (i.e. spring and winter wheat).

The late vegetative through to reproductive crop growth stages are generally the most sensitive periods for crop development.

Conditions:**Drivers:****Rice**

In **China**, harvest of late rice is finished under favourable conditions. In **India**, conditions are favourable Kharif rice as harvest is complete in the northern states and beginning in the southern and eastern states. Sowing of Rabi rice has begun and progressing well under favourable conditions. In **Indonesia**, harvest of dry-season rice is almost complete with yields remaining above last year's. Sowing of wet-season rice is advancing under favourable conditions, with rainfall in late October and mid-November improving conditions. In **Viet Nam**, harvest of the summer-autumn rice (wet-season rice) is complete with yields slightly above last year's. In **Thailand**, conditions have deteriorated for the wet-season rice in the northeast due to continued flooding. In the north, dry conditions during the grain filling stage have caused damage. In the **Philippines**, harvest of wet-season rice is ongoing for rice sown in July through August. Continued damage from multiple typhoons throughout the season have further downgraded rice conditions. In **Brazil**, conditions are favourable as sowing advances.

Soybeans

In **India**, harvest has completed in the main producing areas under favourable conditions. In **Canada**, harvest is wrapping up with generally favourable yields despite late season adverse weather which delayed harvest. In **Brazil**, sowing is ongoing under favourable conditions in the main producing regions. An increase in total sown area expected this season. In **Argentina**, sowing of the spring-planted crop is progressing under favourable conditions with occasional delays due to heavy rainfall.

Information on crop conditions in non-AMIS countries can be found in the [GEOGLAM Early Warning Crop Monitor](#), published 6 December 2018

Policy developments

Wheat

- On 20 November, **China** lowered the minimum purchasing price for wheat in 2019 to CNY 112 per 50 kg (USD 322.2 per tonne). This represents a 3 percent decrease compared to the 2018 price and is the second year in a row in which China has decreased its price support for wheat growers.

Rice

- The National Food Authority (NFA) Council in the **Philippines** approved the out-quota guidelines that will reduce the requirements for local traders to import rice. From 22 November onwards, interested traders can apply for import permits at the NFA. The new guidelines are meant to increase rice imports and consequently bring down the prices of rice in the country.
- In response to the rising paddy prices, **Republic of Korea** announced in early November that it would release around 50 000 tonnes of rice from its public stocks in order to stabilize prices. However, following protests from farmers, the government has decided to withdraw these plans.
- The Ministry of Commerce in **Thailand** will provide rice farmers with harvesting machines at low cost in order to allow farmers to benefit from the projected higher paddy rice prices next year.

Soybean

- The government of Heilongjiang province in **China**, the largest soybean-producing province, has announced that it increased the soybean subsidy from 173 CNY per mu in 2017 to 320 CNY per mu (USD 280 per acre) in 2018.

Biofuels

- The Ministry of Energy in **Argentina** announced a price increase for maize-based ethanol and biodiesel and a price decrease for sugarcane-based ethanol for fuel companies, starting in November 2018. The prices were set at USD 0.55 per litre for maize-based ethanol (higher by 8 percent), USD 782.3 per metric tonne for biodiesel (higher by 2 percent), and USD 0.59 per litre for cane-based ethanol (lower by 3 percent).
- The National Council for Energy Policies in **Brazil** proposed to increase the biodiesel blending mandate – which is currently 10 percent – by one percentage point per year, starting from June 2019. The increase would continue for the next five years, reaching 15 percent by 2023.

- On 5 November, the Department of Commerce of the **US** has started a “changed circumstances” review process that examines the antidumping and countervailing duties that the US imposed on Argentine biodiesel companies. This review process was requested by the government of Argentina after it had increased its export taxes on biodiesel and raised its biodiesel prices. A “changed circumstances” review usually takes around 270 days, meaning that a final determination could be reached by August 2019.
- On 30 November, the Environmental Protection Agency in the **US**, affirmed a de-facto maize ethanol mandate of 15 billion gallons and expanded the requirement for advanced biofuels by 500 million gallons in 2019.
- On 13 November, the **European** Parliament confirmed the bloc’s plan to phase-out, by 2030, the consumption of crop-based biofuels that bring about strong indirect land use changes.

Logistics/Infrastructure/Trade Junctures

- On 21 November, **Ukraine** national railway operator (Ukrzalyznytsya) imposed temporary restrictions, lasting from 20 to 22/23 November, on grain shipments to the Black Sea ports of Odessa and Mykolaiv.
- The naval incident through the Kerch Strait on November 24, involving the **Russian Federation** and **Ukraine**, set out a shipping standoff between the two countries. While Kerch Strait is an important waterway connecting the Sea of Azov with the Black Sea, the overall impact on grain shipments in the event of further escalation remained limited. Only a small portion (less than 6 percent) of grain exports from Ukraine rely on Azov Sea, whereas exports from the Russian Federation, which relies more significantly on the Strait, is unlikely to be disrupted.

Press Stop

- On 26 October, **Mexico** announced the introduction of guaranteed minimum prices for certain crops. Effective from 1 December, minimum paddy (rice) prices would be set at MXN 6 120 (USD 303) per tonne for purchases up to 120 tonnes and from milling wheat, at MXN 5 790 (USD 286) per tonne, with a purchase limit of 100 tonnes. Starting from 1 January 2019, guaranteed maize price would be set at MXN 5 610 (USD 277) per tonne, with a limit of 20 tonnes for each purchase.



AMIS Policy database

Visit the AMIS Policy database at: <http://statistics.amis-outlook.org/policy/>

The AMIS Policy database gathers information on trade measures and domestic measures related to the four AMIS crops (wheat, maize, rice, and soybeans) as well as biofuels. The design of this database allows comparisons across countries, across commodities and across policies for selected periods of time.

International prices

International Grains Council (IGC) Grains and Oilseeds Index (GOI) and GOI sub-Indices

	Nov 2018 Average*	% Change	
		M/M	Y/Y
GOI	195	- 2.7%	+1.0%
Wheat	197	- 1.6%	+12.5%
Maize	173	- 0.3%	+5.2%
Rice	161	- 1.3%	- 2.7%
Soybeans	182	- 4.4%	- 5.1%

*Jan 2000=100, derived from daily export quotations

Wheat

Global wheat export prices posted net declines during November, weighed by ample nearby supplies, strong competition for exports and weakness in outside markets, especially crude oil. There was some support from concerns about prospects for southern hemisphere crops, as recent unfavourable weather in Argentina added to expectations for a below-average sized harvest in Australia. Adverse dryness and cold for 2019/20 crops in parts of Europe and the CIS also provided some price underpinning, as well as seeding delays in the US.

Maize

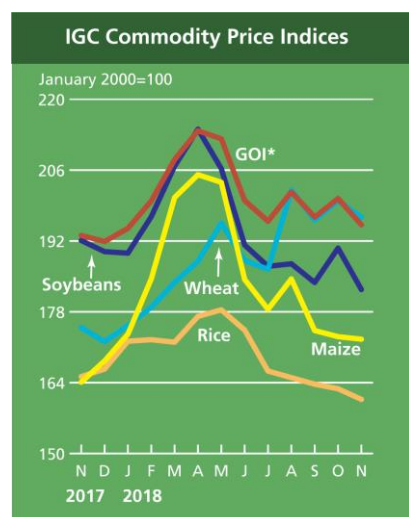
A mostly stable tone was observed in global maize export values in November, with the IGC-GOI sub-Index holding steady for a second successive month. US quotations weakened slightly on ample spot availabilities and signs of stiffer global export competition. After comparatively steep declines since the summer, FOB prices in Ukraine were unchanged m/m, as seasonal harvest pressure was broadly offset by a pickup in overseas demand. Average prices in Brazil also showed little change. However, with supplies beginning to tighten, offers were recently quoted at a modest premium over competing origins.

Rice

Tied to pressure from new crop arrivals in key exporters and generally slow demand, world white and parboiled rice values were mostly lower during November, the IGC-GOI rice sub-Index falling by 1.3 percent m/m. However, overall declines were limited by fresh sales to the Philippines in a recent government tender, filled by supplies from both Vietnam and Thailand. Outside of Asia, US Gulf milled rice prices showed little overall change on support from good export interest.

Soybeans

Average soybean prices retreated by 4.4 percent m/m during November, although trends across key origins were mixed. Despite pressure from sluggish international demand and harvesting of what is set to be the largest crop on record, US Gulf values strengthened on increasing hopes for a breakthrough in the trade dispute with China. In contrast, southern hemisphere markets were significantly weaker. In Brazil, the record planting pace weighed on sentiment, as did an erosion of spot buying interest against the backdrop of reported trade discussions between the US and China. In Argentina, too, where seeding was close to the halfway stage, export prices were lower.

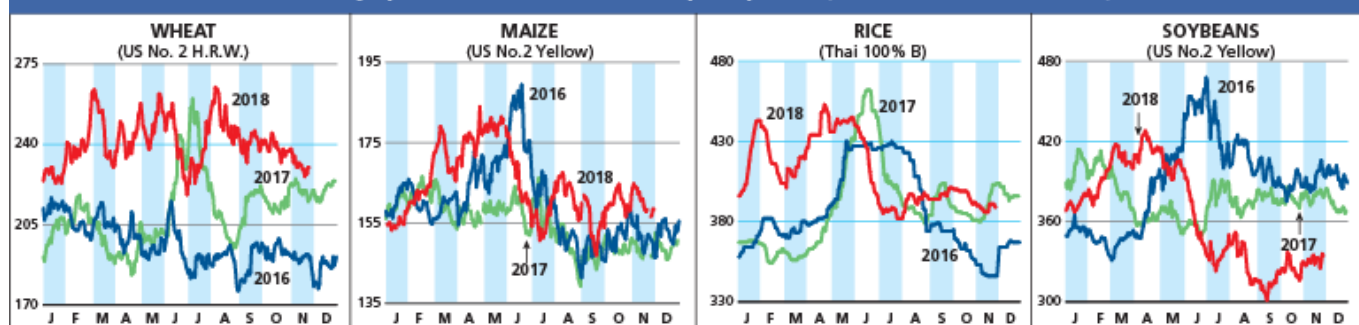


		IGC commodity price indices				
		GOI*	Wheat	Maize	Rice	Soybeans
		(..... January 2000 = 100)				
2017	November	193.1	174.9	164.1	165.2	192.1
	December	191.9	172.1	168.3	166.7	189.9
2018	January	194.5	175.3	173.6	172.2	189.6
	February	199.9	178.9	184.5	172.5	196.8
	March	208.1	183.8	200.6	172.0	206.8
	April	213.8	188.0	205.1	177.1	214.1
	May	212.2	195.5	203.6	178.4	206.3
	June	200.0	188.2	184.4	174.4	191.2
	July	195.9	186.4	178.5	166.3	187.0
	August	201.7	202.1	184.5	165.0	187.5
	September	196.7	196.1	174.3	163.7	183.8
	October	200.4	200.0	173.1	162.8	190.6
	November	195.2	196.7	172.6	160.7	182.4

*GOI: Grains and Oilseeds Index

Selected export prices, currencies and indices

Daily quotations of selected export prices (USD/tonne, 2016-2018)



Daily quotations of selected export prices

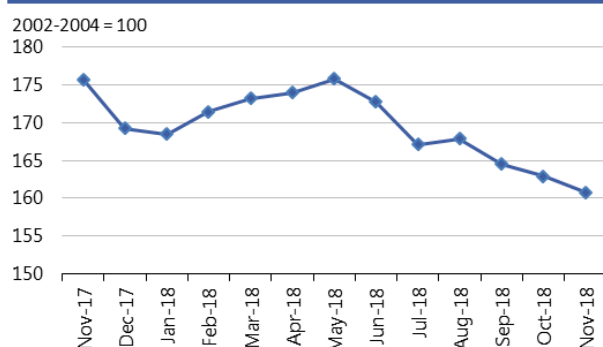
	Effective Date	Quotation (1)	Week ago (2)	Month ago (3)	Year ago (4)	% change (1) over (2)	% change (1) over (4)
(..... USD/tonne) (.....)							
Wheat (US No. 2, HRW)	29-Nov	230	229	238	217	0.4%	6.0%
Maize (US No. 2, Yellow)	29-Nov	158	158	159	147	0.0%	7.4%
Rice (Thai 100% B)	29-Nov	389	391	393	403	-0.5%	-3.5%
Soybeans (US No. 2, Yellow)	29-Nov	334	333	318	381	0.3%	-12.3%

AMIS Countries' Currencies Against US Dollar

AMIS Countries	Currency	Nov 2018 Average	Monthly Change	Annual Change
Argentina	ARS	36.4	1.7%	-108.3%
Australia	AUD	1.4	2.0%	-5.1%
Brazil	BRL	3.8	-0.5%	-16.0%
Canada	CAD	1.3	-1.5%	-3.4%
China	CNY	6.9	-0.3%	-4.8%
Egypt	EGP	17.9	0.0%	-1.3%
EU	EUR	0.9	-1.0%	-3.3%
India	INR	71.7	2.5%	-10.6%
Indonesia	IDR	14,655.0	3.4%	-8.4%
Japan	JPY	113.4	-0.5%	-0.5%
Kazakhstan	KZT	370.8	-0.9%	-11.6%
Rep. Korea	KRW	1,125.4	0.6%	-2.2%
Mexico	MXN	20.2	-5.3%	-7.0%
Nigeria	NGN	305.7	-0.1%	0.1%
Philippines	PHP	52.7	2.2%	-3.6%
Russian Fed.	RUB	66.5	-1.0%	-12.8%
Saudi Arabia	SAR	3.8	0.0%	0.0%
South Africa	ZAR	14.1	3.2%	-0.1%
Thailand	THB	32.9	-0.5%	-0.2%
Turkey	TRY	5.4	7.1%	-39.2%
UK	GBP	0.8	-0.8%	-2.6%
Ukraine	UAH	27.9	0.7%	-4.5%
Viet Nam	VND	23,324.8	0.1%	-2.7%

FAO Food Price Index

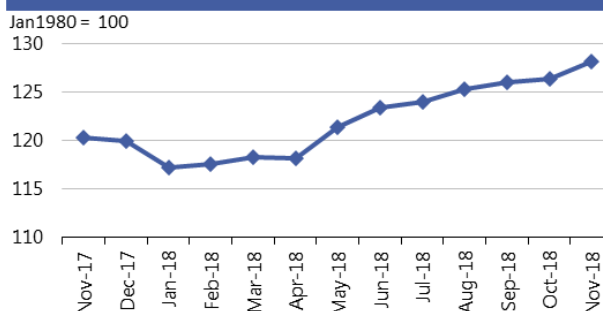
Nov 2017-Nov 2018



Source: FAO

Nominal Broad Dollar Index

Nov 2017-Nov 2018



Source: US Federal Reserve

Futures markets

Futures Prices – nearby

	Nov-18 Average	% Change	
		M/M	Y/Y
Wheat	186	- 1.2%	+19.7%
Maize	144	- 0.6%	+ 6.6%
Rice	237	+ 0.3%	- 8.7%
Soybeans	322	+2.0%	- 10.8%

Source: CME

Futures Prices

Prices for wheat, maize, soybeans and rice showed negligible change m/m, with soybeans rising by 2.0 percent, wheat falling by 1.2 percent and maize and rice changing little. Except for a brief advance during the first week of November when adverse weather threatened harvest prospects, prices for wheat, maize and soybeans drifted lower. The protracted stalemate over Chinese tariffs on US soybeans, as well as the effects of exogenous markets - especially energy - added further pressure to values. After reaching a four-year high in early October, crude oil plummeted by one-third from USD 76 to USD 51 per barrel, negatively impacting the demand for maize-based ethanol. Currency appreciation may have also added to bearish commodity sentiment as the USD index reached an 18 month high. Reflecting their divergent supply and demand fundamentals y/y, wheat and maize were higher by 20 and 7 percent respectively, while soybeans and rice were lower by 11 and 9 percent respectively.

Volumes and volatility

Trade volumes for wheat and maize rose nearly 50 percent m/m but fell for soybeans by 38 percent, while volumes for all three commodities were lower marginally y/y. Implied volatility and historical volatility fell for wheat and maize by varying degrees m/m while rising slightly for soybeans. Both volatility measures were higher for all three commodities y/y than the 2017 ultra-low levels.

Basis levels and transport

Domestic basis levels firmed in the interior m/m for maize and soybeans, as a bumper harvest neared completion for the two commodities. In Illinois, the interior bids to local elevators were quoted at minus USD 11 per tonne for maize and minus USD 18 per tonne for soybeans, each under the respective December and January futures prices. In Iowa, the bids were minus USD 14 for maize and minus USD 33 for soybeans (under the respective futures). In soft red wheat, bids for delivery to northern mills were quoted at premiums to December futures prices. Gulf export delivery basis levels for maize, soybeans and wheat were about unchanged m/m, at USD 16, USD 7, and USD 22 (per tonne premium over respective futures). Barge freight (lower Illinois River quotations) dropped precipitously m/m from USD 26 to USD 16 per tonne, signaling the completion of harvest. In the export market, total export commitments (shipped and

Historical Volatility – 30 Days, nearby

	Monthly Averages		
	Nov-18	Oct-18	Nov-17
Wheat	21.9	27.9	17.9
Maize	15.6	22.5	12.6
Rice	18.3	24.1	21.5
Soybeans	19.1	17.5	13.3

unshipped) for maize remained robust, but lagged behind in wheat on a crop y/y comparison. Soybeans experienced a steep drop in total export commitments of 10million tonnes since the start of the September 1 crop year, reflecting a 32 percent decline due to halted shipments to China.

Forward curves

Forward curves exhibited slight changes m/m, decreasing in slope (narrowing) for wheat and widening for maize, with all curves remaining in contango configurations (upward sloping). Analysts attributed the narrowing of the y/y wheat curves (between December 18 and December 19) from minus USD 25 to USD 15 per tonne to strong basis levels and the prospect of increased exports during the second half of the crop year, particularly following a recent US gulf sale to Egypt. The widening of the y/y maize curve from minus USD 12 to USD 14 per tonne was most likely a result of the downward trend of the nearby futures. The y/y soybean spread that ended with the mid-month expiration of the November 2018 contract, expired at the widest negative level recorded for any November/November spread at minus USD 24, testifying to the gloomy export demand level and large global supplies.

Investment flows

Managed money made modest changes to its positions m/m, decreasing its net short in wheat, increasing its net short in soybeans while switching its net long to net short in maize. Its change in trade strategies y/y from large directional bets to more sophisticated spread bets which depend on the convergence/divergence of the price differential between two contract months (e.g. March and May) is most evident in maize. Whereas, during November 2017, managed money held a net short of almost 200 000 contracts, it held a net short of only 10 000 contracts according to the most recent data. Simultaneously, managed money increased its spread positions y/y from 120 000 to 222 000 contracts. Commercials remained net short m/m in all three commodities, while swaps dealers remained the dominant net long.

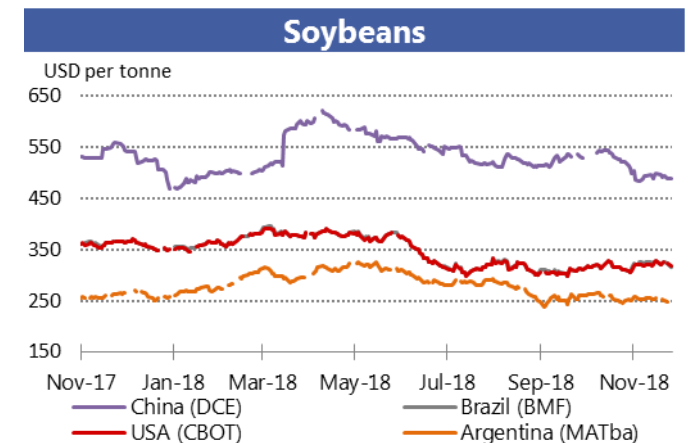
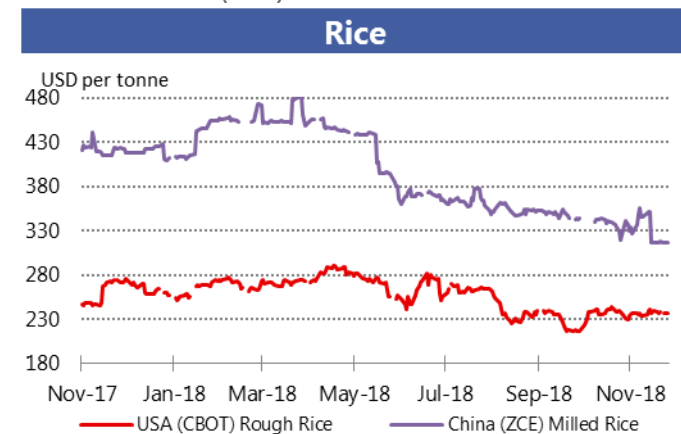
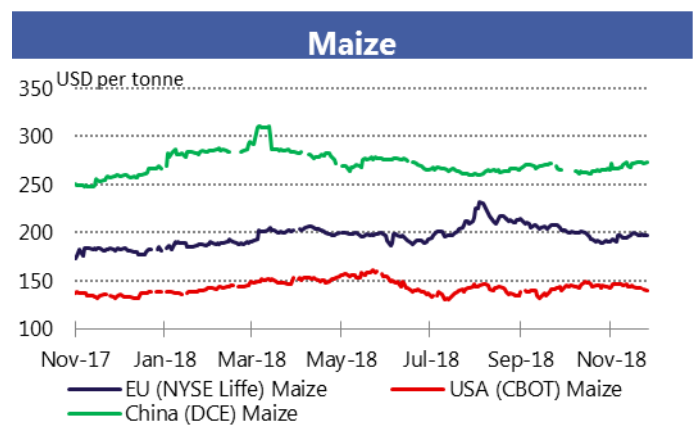
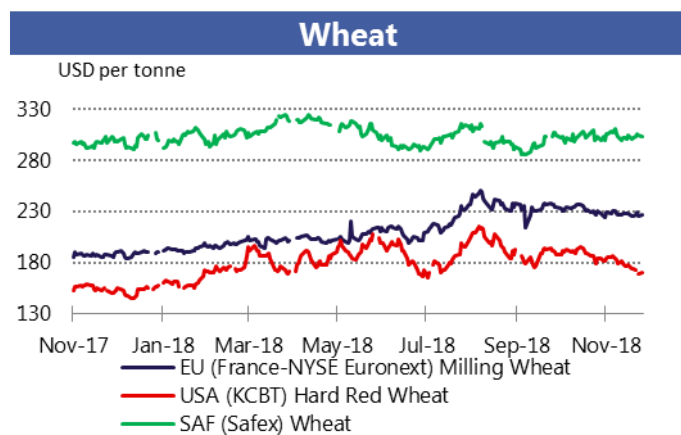


For more information on technical terms please see the Glossary at the following link:

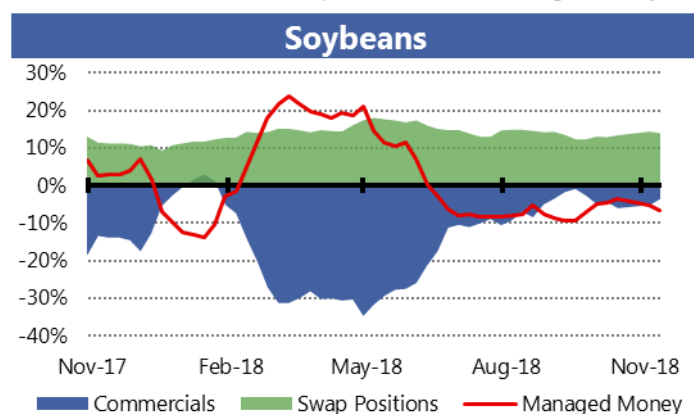
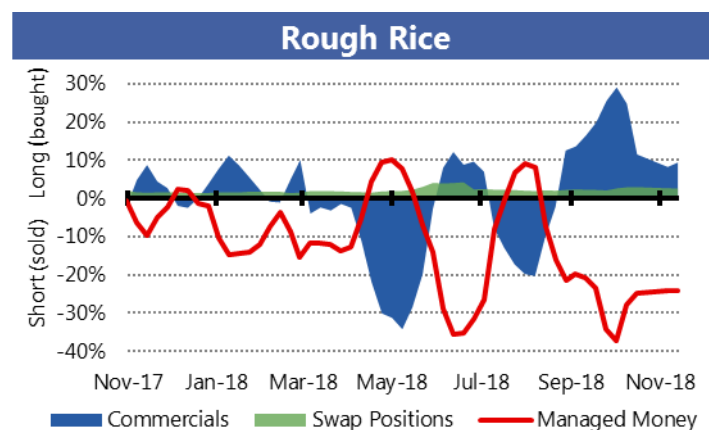
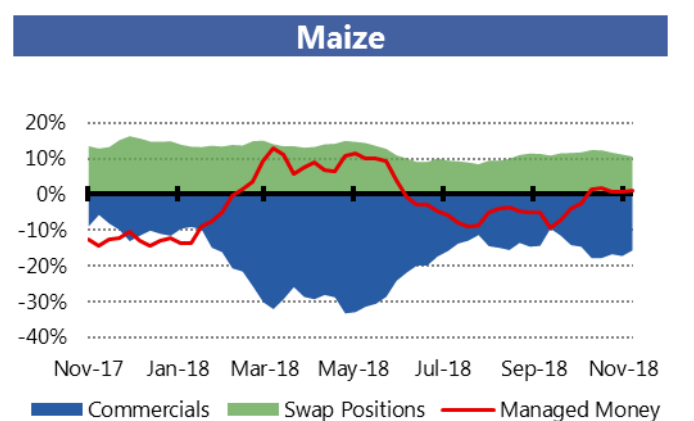
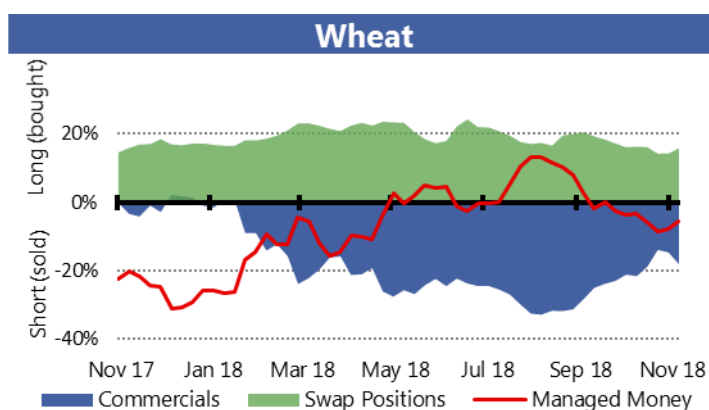
http://www.amis-outlook.org/fileadmin/user_upload/amis/docs/Market_monitor/Glossary.pdf

Market indicators

Daily quotations from leading exchanges - nearby futures

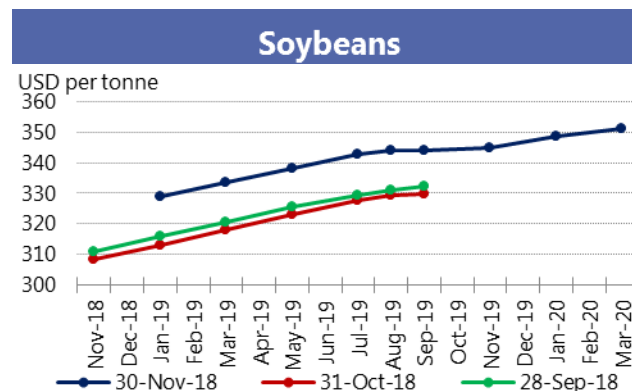
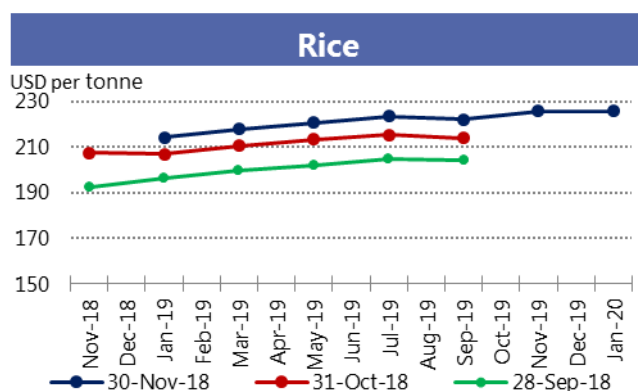
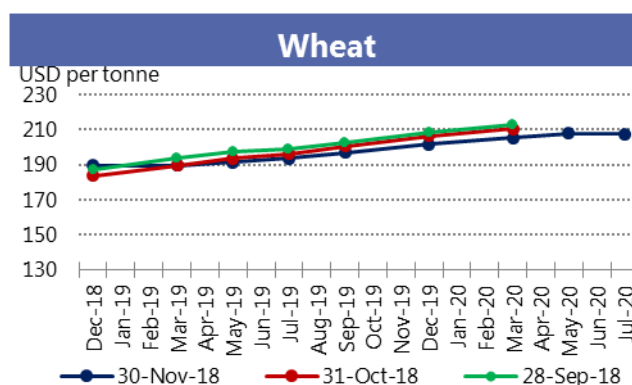
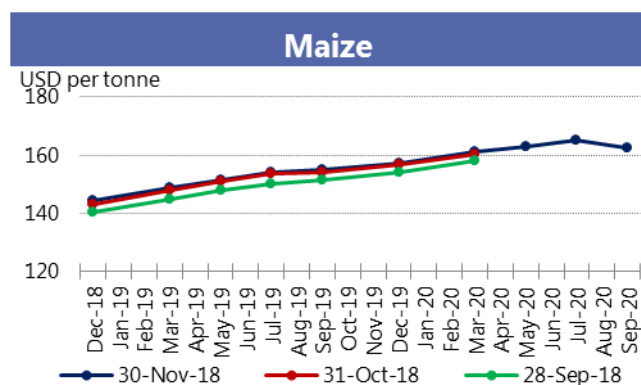


CFTC Commitments of Traders - Major Categories Net Length as percentage of Open Interest*

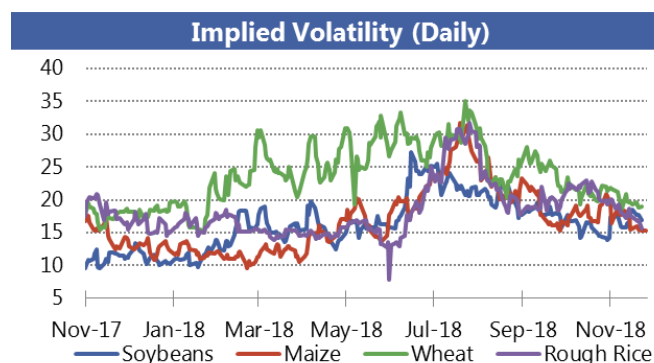
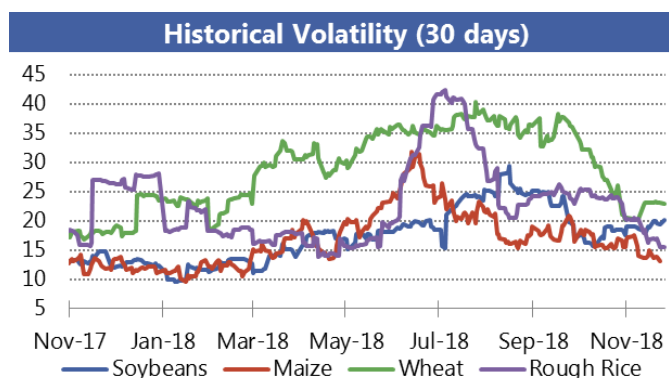


*Disaggregated Futures Only. Though not all positions are reflected in the charts, total long positions always equal total short positions.

Forward Curves



Historical and Implied Volatilities



i AMIS Market indicators

Some of the indicators covered in this report are updated regularly on the AMIS website. These, as well as other market indicators, can be found at: <http://www.amis-outlook.org/amis-monitoring/indicators/>

Monthly US ethanol update

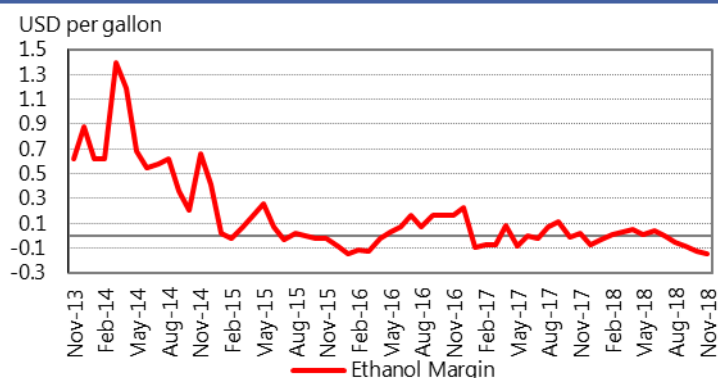
- **Ethanol production margins** remain negative, and significantly so for the second consecutive month.
- Rising **maize prices** were the primary driver for falling margins.
- **Oil and gasoline prices** fell precipitously through November while **ethanol prices** remained relatively stable, reducing the notable price advantage that ethanol held in September and into early October.
- The pace of **ethanol production** rebounded in November at an annualized pace of 16.275 billion gallons.
- On 30 November, the Environmental Protection Agency in the US, affirmed a de-facto maize **ethanol mandate** of 15 billion gallons and expanded the requirement for advanced biofuels by 500 million gallons in 2019.

Spot prices	Nov	Oct	Nov
IA, NE and IL/eastern corn belt average	2018*	2018	2017
Maize price (USD per tonne)	133.70	131.05	125.69
DDGs (USD per tonne)	136.45	135.42	122.65
Ethanol price (USD per gallon)	1.22	1.22	1.35
Nearby futures prices			
CME, NYSE			
Ethanol (USD per gallon)	1.26	1.29	1.42
RBOB Gasoline (USD per gallon)	1.63	1.93	1.77
Ethanol/RBOB price ratio	77.5%	66.6%	80.2%
Ethanol margins			
IA, NE and IL/eastern corn belt			
Average (USD per gallon)			
Ethanol receipts	1.22	1.22	1.35
DDGs receipts	0.42	0.42	0.38
Maize costs	1.24	1.21	1.16
Other costs	0.55	0.55	0.55
Production margin	-0.14	-0.13	0.01
Ethanol production			
(million gallons)			
Monthly production total	1 338	1 362	1 364
Annualized production pace	16 275	16 042	16 592

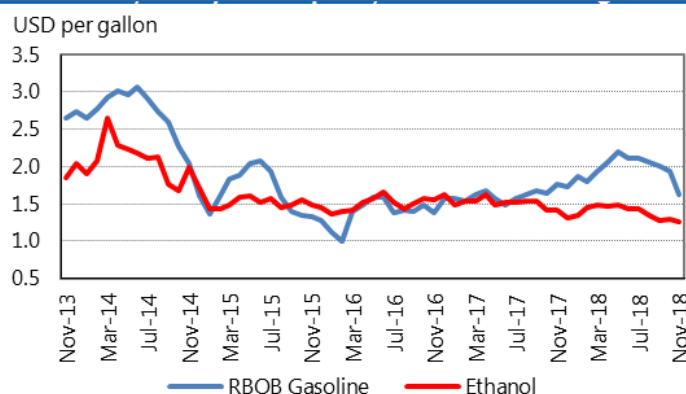
Based on USDA data and private sources

* Estimated using available weekly data to date.

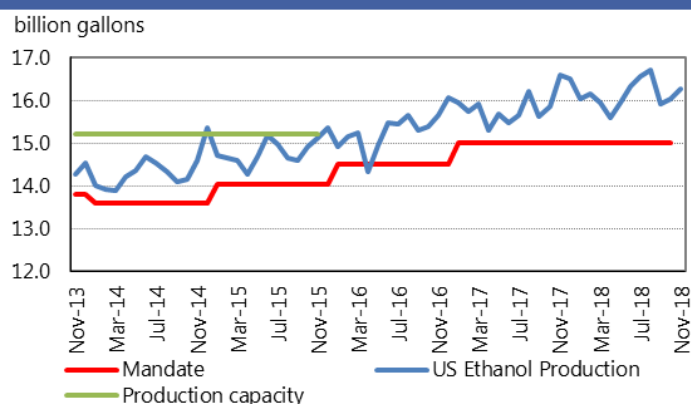
Ethanol Production Margin
(IA, NE, IL/eastern corn belt average)



Ethanol and RBOB gasoline
(nearby futures prices, CME, NYSE)



Ethanol production pace, capacity and annual mandate



Ethanol price vs. maize price
(Spot prices)

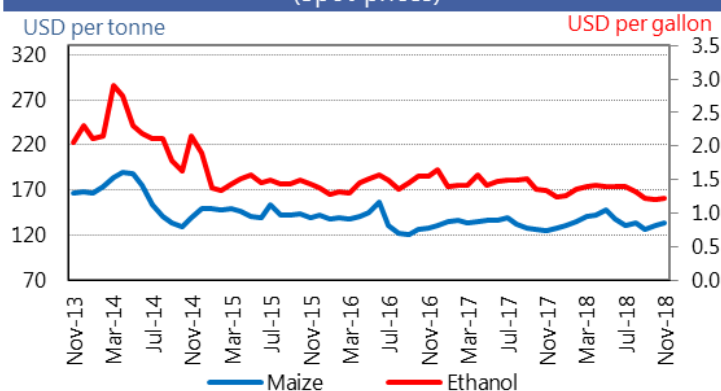


Chart and tables description

Ethanol Production Margins: The ethanol margin gives an indication of the profitability of maize-based ethanol production in the United States. It uses current market prices for maize, Dried Distillers Grains (DDGs) and ethanol, with an additional USD 0.55 per gallon of production costs

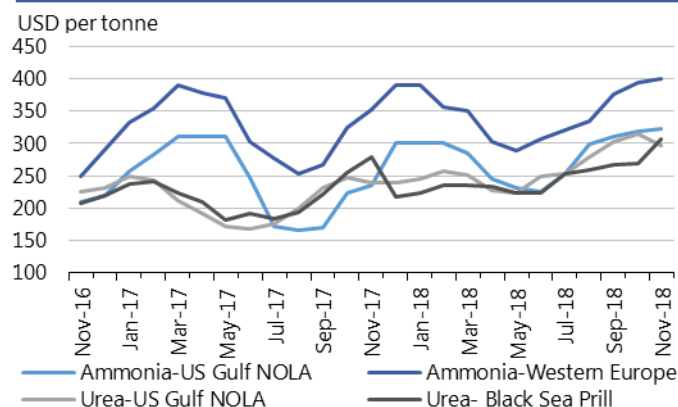
Ethanol Production Pace, Capacity and Mandate: Overview of the volume of maize-based ethanol production in the United States; it also highlights overall production capacity and the production volume that is mandated by public legislation. Name-plate (i.e. nominal) ethanol production capacity in the US is roughly 14.9 billion gallons per annum, but plants can exceed this level, so the actual capacity is assumed to be 15.2 billion gallons.

DDGs: By-product of maize-based biofuel production, commonly used as feedstuff.

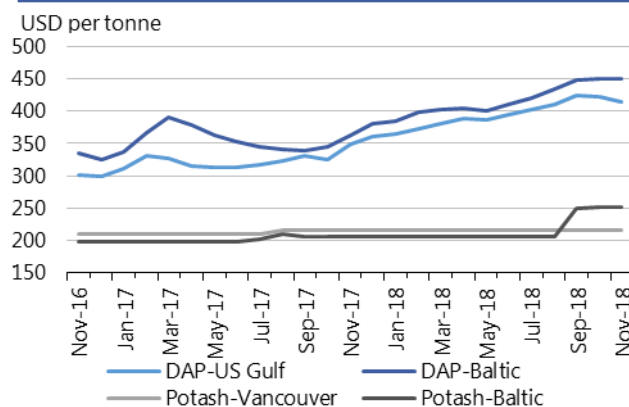
RBOB: Reformulated Blendstock for Oxygenate Blending, gasoline nearby futures (NYSE).

Fertilizer outlook

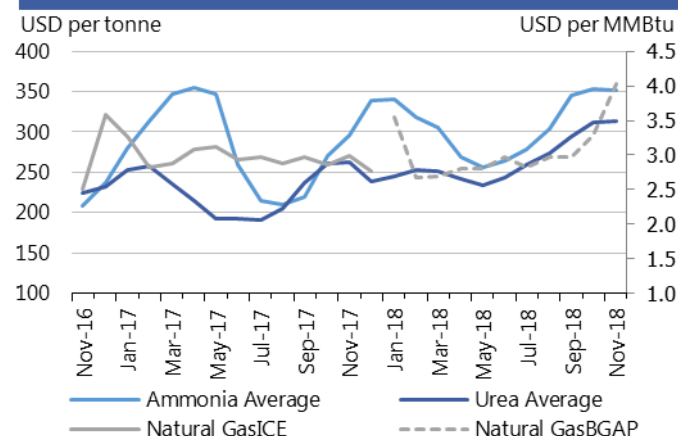
Ammonia and Urea
(Spot prices)



Potash and Phosphate
(Spot prices)



Ammonia Average, Urea Average and Natural Gas
(Spot prices)



- **Natural gas** prices spiked in November due to cold temperatures and low inventories in the US.
- **Ammonia** prices continued to increase, but at a slower rate than previous months due to the end of seasonal demand in India, which relieved upward pressure on prices.
- **Urea** prices showed a significant increase in the Black Sea due to supply restrictions in China. In contrast, prices decreased in the US Gulf due to delayed applications caused by unfavourable weather in the US.
- **DAP** prices have stabilized in response to rising input (nitrogen) prices in India (pushing prices upward), combined with a delayed application in the US due to bad weather conditions (pushing prices downward).
- **Potash** prices in North America remained unchanged m/m, reflecting the delayed fall application season in the US.

	November average	November std. dev	% change last month*	% change last year*	12-month high	12-month low
Ammonia-US Gulf NOLA	322.0	-	0.9%	37.0%	322.0	226.0
Ammonia-Western Europe	400.0	11.5	1.3%	13.5%	400.0	289.0
Urea-US Gulf	296.8	5.6	-6.0%	24.6%	315.8	224.0
Urea-Black Sea	306.0	4.0	13.3%	9.3%	306.0	217.5
DAP-US Gulf	414.0	4.5	-1.8%	19.0%	423.8	360.0
DAP-Baltic	450.0	-	0.0%	24.1%	450.0	380.0
Potash-Baltic	252.0	-	0.0%	22.3%	252.0	206.0
Potash-Vancouver	216.0	-	0.0%	0.0%	216.0	216.0
Ammonia	351.8	7.2	-0.3%	18.9%	352.9	256.3
Urea	313.3	0.8	0.2%	19.0%	313.3	234.0
Natural Gas*	4.0	0.6	22.5%	35.9%	4.0	2.7

All prices shown are in US dollars.

*Natural Gas is a new Henry Hub Index (BGAP), replacing the one used before, which has been discontinued.

Source: Own elaboration based on Bloomberg



Chart and tables description

Ammonia and Urea: Overview of nitrogen-based fertilizer prices in the US Gulf, Western Europe and Black Sea. Prices are weekly prices averaged by month.

Potash and Phosphate: Overview of phosphate and potassium-based fertilizer prices in the US Gulf, Baltic and Vancouver. Prices are weekly prices averaged by month.

Ammonia Average and Urea Average: Monthly average prices from Ammonia's US Gulf NOLA, Middle East, Black Sea and Western Europe were averaged to obtain Ammonia Average prices; monthly average prices from Urea's US Gulf NOLA, US Gulf Prill, Middle East Prill, Black Sea Prill and Mediterranean were averaged to obtain Urea Average prices.

Natural Gas: Henry Hub Natural Gas Spot Price from ICE up to December 2017 and from Bloomberg (BGAP) from January 2018 onwards. Prices are intraday prices averaged by month. Natural gas is used as major input to produce nitrogen-based fertilizers.

DAP: Diammonium Phosphate.

Monthly ocean freight market update

Dry bulk freight market developments

	Nov 2018 Average*	M/M	% Change
Baltic Dry Index (BDI) *	1 186	- 23.2%	- 18.5%
<i>sub-Indices:</i>			
Capesize	1 605	- 28.2%	- 51.4%
Panamax	1 463	-14.0%	+ 5.2%
Supramax	988	- 15.6%	+ 6.9%
Baltic Handysize Index (BHSI)**	649	- 1.2%	+ 1.6%

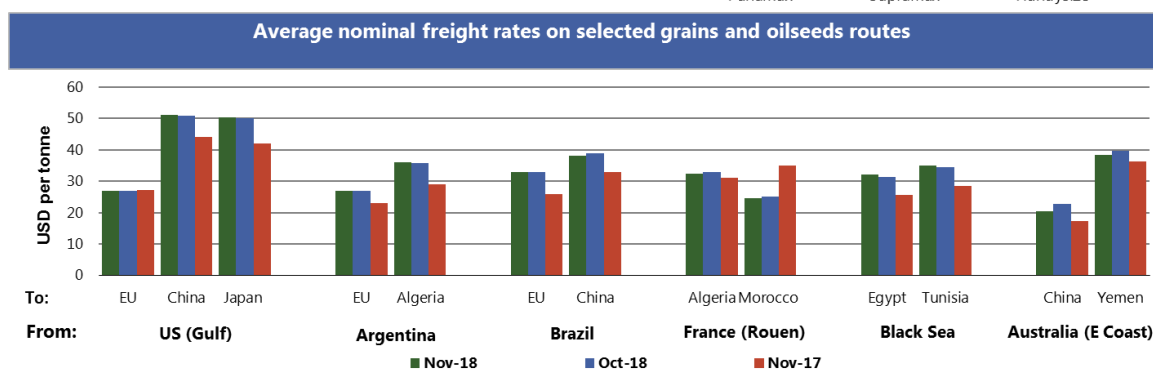
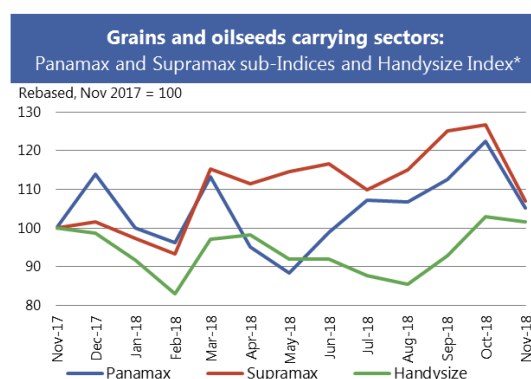
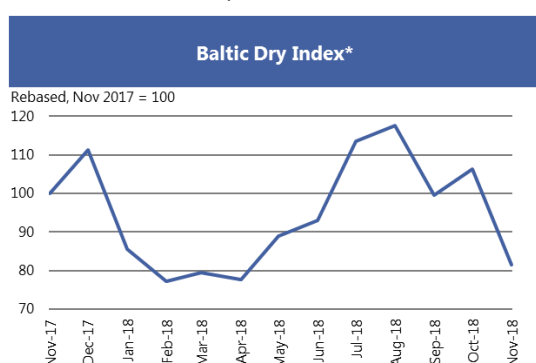
Source: Baltic Exchange.

Note: *4 January 1985 = 1000 **23 May 2006 = 1000. Baltic Handysize sub-Index excluded from the BDI from 1 March 2018

- With market direction often shaped by gyrations in the non-grains and oilseeds related Capesize market, average **Baltic Dry Index (BDI)** quotations declined by almost one-quarter in November, to the lowest since April 2018. A softer tone in the grains and oilseeds carrying segments added to losses, with the BDI down 19 percent y/y.
- Despite a recent rally, **Capesize** values were down by 28 percent on average amid reduced enquiries and ample tonnage supply on leading routes. Sentiment was weighed by reports of poor steel mill margins in China and talk of fresh anti-pollution measures there.
- Partly linked to spillover from the Capesize segment, average **Panamax** earnings eased by 14 percent m/m. Weakness was evident in the Pacific, where pressure from reduced demand was amplified by speculation about new import restrictions on coal shipments to China. In contrast,

good volumes were reported in the Atlantic, notably for coal and maize dispatches from the US Gulf. A strong end to Brazil's soyabean export season buoyed South American levels, but an influx of vessels from other areas limited upside.

- Demand in the **Supramax** and **Handysize** segments also bucked expectations as Baltic sub-Indices fell by 16 percent and 1 percent, respectively. The former saw erratic demand in the Pacific, coupled with discounting in Europe. While this was partly offset by sustained interest for grains shipments from the Black Sea region, observers noted uncertain prospects, tied to tensions between Russian Federation and Ukraine. Fresh demand was also noted in South America. However, building vessel queues off the coast of Argentina following large soyabean arrivals from the US were a recent feature.



* monthly average. Source: International Grains Council

Notes:

Baltic Dry Index (BDI): A global benchmark indicator issued daily by the London-based Baltic Exchange, providing an assessment of the costs of moving major raw materials on ocean going vessels. The BDI is a composite measure, comprising sub-indices for four carrying segments, representing different vessel sizes: Capesize, Panamax, Supramax and Handysize. **Capesize:** The largest vessels included in the BDI with deadweight tonnage (DWT) above 80 000 DWT, primarily transporting coal, iron ore and other heavy raw materials on long-haul routes.

Panamax: Vessels with capacity of 60 000 to 80 000 DWT, which are mostly geared to transporting coal, grains, oilseeds and other bulks, including sugar and cement.

Supramax/Handysize: Vessels with capacity below 60 000 DWT, which account for the majority of the world's ocean going vessels. They can transport a wide variety of cargos, including grains and oilseeds.

Explanatory Notes

The notions of **tightening** and **easing** used in the summary table of “Markets at a glance” reflect judgmental views that take into account market fundamentals, inter-alia price developments and short-term trends in demand and supply, especially changes in stocks.

All totals (aggregates) are computed from unrounded data. World supply and demand estimates/forecasts are based on the latest data published by FAO, IGC and USDA. For the former, they also take into account information provided by AMIS focal points (hence the notion “FAO-AMIS”). World estimates and forecasts produced by the three sources may vary due to several reasons, such as varying release dates and different methodologies used in constructing commodity balances. Specifically:

Production: Wheat production data from all three sources refer to production occurring in the first year of the marketing season shown (e.g. crops harvested in 2016 are allocated to the 2016/17 marketing season). Maize and rice production data for FAO-AMIS refer to crops harvested during the first year of the marketing season (e.g. 2016 for the 2016/17 marketing season) in both the northern and southern hemisphere. Rice production data for FAO-AMIS also include northern hemisphere production from secondary crops harvested in the second year of the marketing season (e.g. 2017 for the 2016/17 marketing season). By contrast, rice and maize data for USDA and IGC encompass production in the northern hemisphere occurring during the first year of the season (e.g. 2016 for the 2016/17 marketing season), as well as crops harvested in the southern hemisphere during the second year of the season (e.g. 2017 for the 2016/17 marketing season). For soybeans, the latter approach is used by all three sources.

Supply: Defined as production plus opening stocks by all three sources.

Utilization: For all three sources, wheat, maize and rice utilization includes food, feed and other uses (namely, seeds, industrial uses and post-harvest losses). For soybeans, it comprises crush, food and other uses. However, for all AMIS commodities, the use categories may be grouped differently across sources and may also include residual values.

Trade: Data refer to exports. For wheat and maize, trade is reported on a July/June basis, except for USDA maize trade estimates, which are reported on an October/September basis. Wheat trade data from all three sources includes wheat flour in wheat grain equivalent, while the USDA also considers wheat products. For rice, trade covers shipments from January to December of the second year of the respective marketing season. For soybeans, trade is reported on an October/September basis by FAO-AMIS and the IGC, while USDA data are based on local marketing years except for Argentina and Brazil which are reported on an October/September basis. Trade between European Union member states is excluded.

Stocks: In general, world stocks of AMIS crops refer to the sum of carry-overs at the close of each country's national marketing year. For soybeans, stock levels reported by the USDA are based on local marketing years, except for Argentina and Brazil, which are adjusted to October/September. For maize and rice, global estimates may vary across sources because of differences in the allocation of production in southern hemisphere countries.

For more information on AMIS Supply and Demand, please view [AMIS Supply and Demand Balances Manual](#).

Main sources

Bloomberg, CFTC, CME Group, FAO, GEOGLAM, IFPRI, IGC, Reuters, USDA, US Federal Reserve

AMIS - GEOGLAM Crop Calendar

Selected leading producers

Wheat		J	F	M	A	M	J	J	A	S	O	N	D
EU (21%)*	winter												
	spring												
China (17%)	winter												
India (13%)	winter												
US (8%)	spring												
	winter												
Russia (8%)	spring												
	winter												
Maize		J	F	M	A	M	J	J	A	S	O	N	D
US (35%)													
China (22%)	north												
	south												
Brazil (8%)	1st crop												
	2nd crop												
EU (7%)													
Argentina (3%)													
Rice		J	F	M	A	M	J	J	A	S	O	N	D
China (29%)	intermediary crop												
	late crop												
	early crop												
India (21%)	kharif												
	rabi												
Indonesia (9%)	main Java												
	second Java												
	winter-spring												
Viet Nam (6%)	summer/autumn												
	winter												
Thailand (4%)	main season												
	second season												
Soybeans		J	F	M	A	M	J	J	A	S	O	N	D
USA (31%)													
Brazil (29%)													
Argentina (18%)													
China (4%)													
India (3%)													

* Percentages refer to the global share of production (average 2013-15).

	Planting (peak)		Harvest (peak)
	Planting		Harvest
	Weather conditions in this period are critical for yields.		Growing period

2019 AMIS Market Monitor Release Dates

February 7, March 7, April 4, May 9, June 6, July 4, September 5, October 3, November 7, December 5

Contacts and Subscriptions

AMIS Secretariat Email:

AMIS-Secretariat@fao.org

Download the AMIS Market Monitor or sign up for a free e-mail subscription at:

www.amis-outlook.org/amis-monitoring

